

AS level economics

Chp: Demand and Supply

Topic:- Cross Elasticity of demand

alpha

Farah
Nushtaq

A stylized handwritten signature in black ink, featuring a large loop and a cross-like shape at the top.

Cross Elasticity of demand (XED) $\Rightarrow \frac{\% \Delta \text{ in QD of good Y}}{\% \Delta \text{ in price of good X}}$

measures
responsiveness

+ve, -ve, 0

Substitutes

complements

Unrelated goods

2 goods

Substitutes

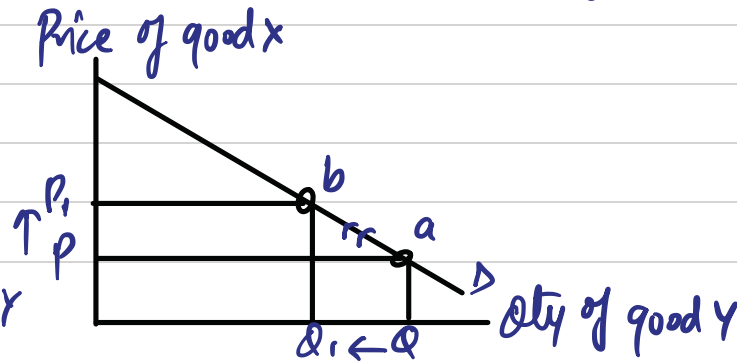
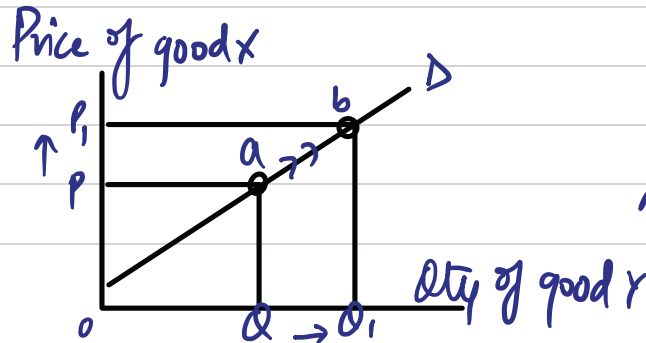
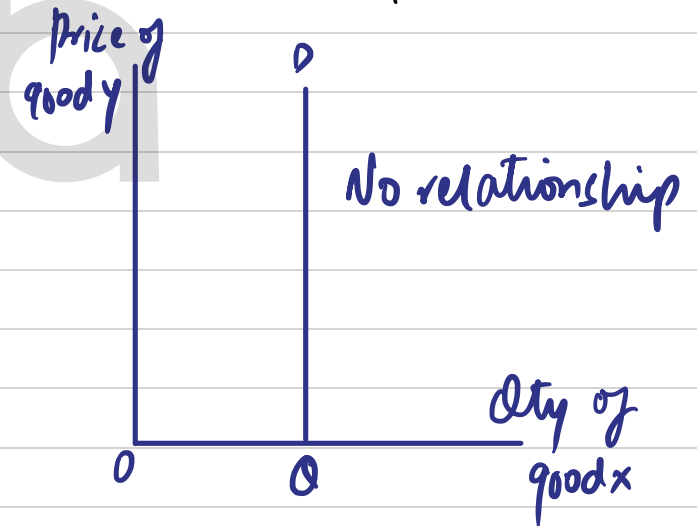
Complements

Coke
Price $\uparrow$ QD $\downarrow$
Price $\downarrow$ QD $\uparrow$
+ve Relationship

pepsi
Demand $\uparrow$
Demand $\downarrow$

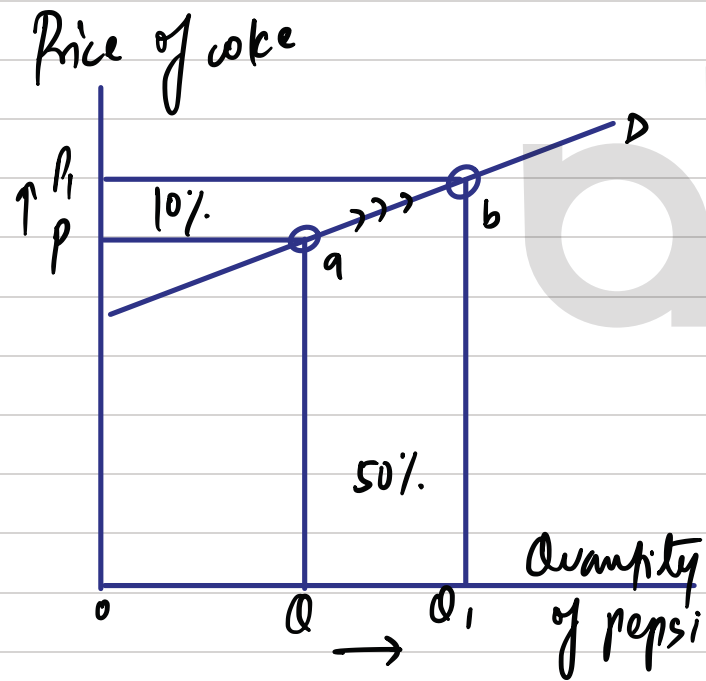
Car
Price $\uparrow$ QD $\downarrow$
Price $\downarrow$ QD $\uparrow$
-ve Relationship

petrol
Demand $\downarrow$
Demand $\uparrow$



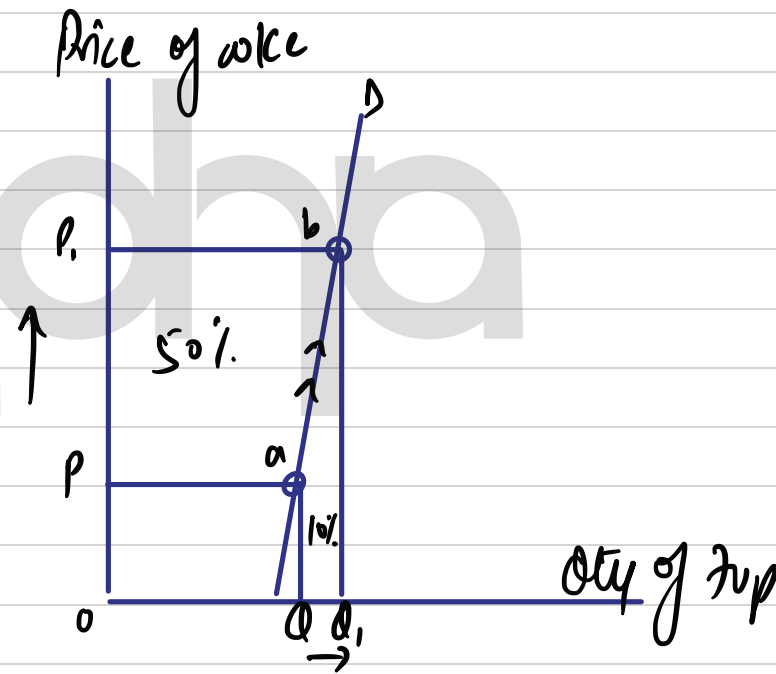
Close and weak substitutes $\Rightarrow$ Positive Relationship
 between price of one good and
 quantity demanded for other good.

e.g. coke and pepsi coke and 7up



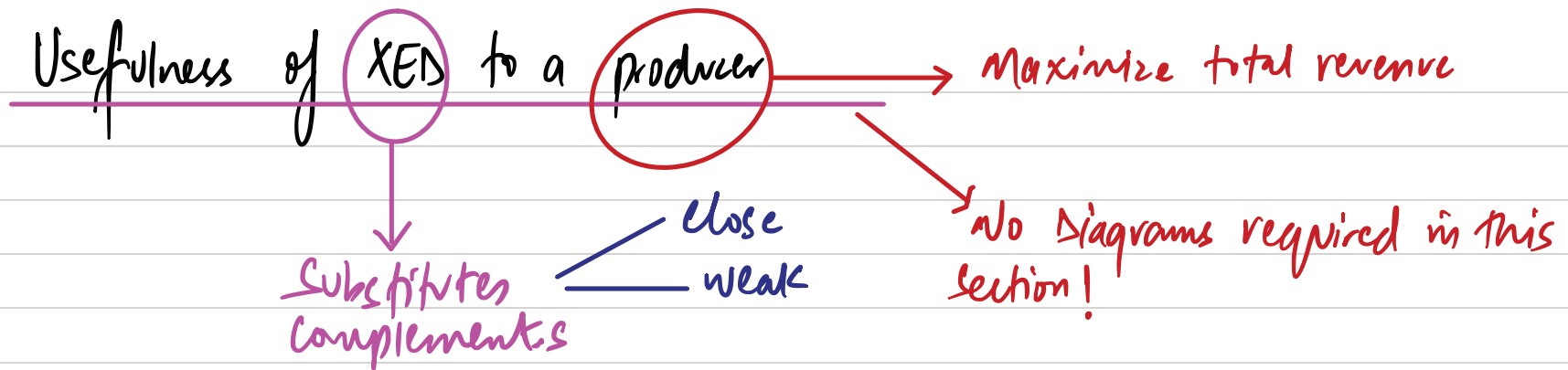
$$XED = \frac{50\%}{10\%} \Rightarrow +5 (>1)$$

↓
Highly elastic



$$XED = \frac{10\%}{50\%} \Rightarrow +0.2 (<1)$$

↓
Inelastic



Strategy :-

Close substitutes, strong rivals, reduce the price by smaller percentage, along with that work on persuasive advertising to change the elasticity, no diagram!

weak substitutes, no close substitutes, prices $\uparrow$ $Q_d \downarrow$, $TR \uparrow$

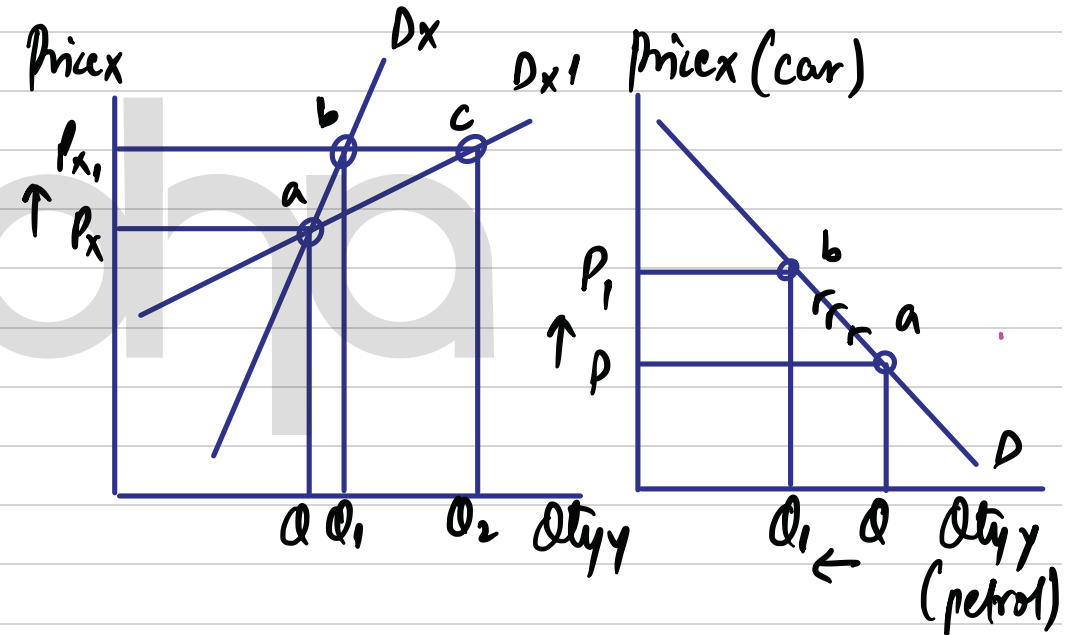
Complements, cinema tickets and popcorn/drinks

$\downarrow$
 Lower the price
 of main product
 $\downarrow$
 Prices $\downarrow$ $Q_d \uparrow$

$\downarrow$
 High profit
 margin on these
 products

Q1 (a) Explain with the help of a diagram, what happens in the market for a product when the price of a substitute increases, and how economists would measure the relationship between these two products ? (8)

- Ans:-
- Define XED
 - formula
 - values +ve, -ve
 - substitutes, complements
 - Define
 - close and weak substitutes
 - Diagram Explanation
 - Stronger the relationship, closer the substitutes



XED :- $\frac{\text{percentage change in Qty demand of good Y}}{\text{percentage change in price of good X}} \Rightarrow \begin{matrix} +ve \\ -ve \end{matrix}$

Q1(b) Discuss whether Income elasticity of demand is more useful than cross elasticity of demand to maximize total revenue? 12

Ans:- 4 marks (Usefulness of YED to a producer)

4 marks (Usefulness of XED to a producer)

4 marks (Which one Better)



Pick XED (Prices involved)
 mention disadvantages of YED! (same covered under YED topic)

→ XED (prices are in control of the producer)

$XED > YED$!

"We don't draw relationship diagram of YED and XED under TR Qs, as the diagrams doesn't reflect TR!"